

Company Choice: BMW (Bayerische Motoren Werke AG)

We selected **BMW** as the car manufacturer for this analysis because it represents a financially strong and strategically attractive company with long-term investment appeal. The company operates in the premium automotive segment, which supports strong brand value, customer loyalty, and pricing power across global markets. From our analysis, BMW shows solid financial performance, efficient operations, and a balanced risk profile, making it a reliable investment candidate. In addition, the BMW brand remains highly demanded in the market, supported by continuous innovation and a clear focus on future mobility.

Selected Peer Group:

BMW, Mercedes-Benz Group AG and Volkswagen AG were selected as the peer group for this analysis. They are relevant because they are major European car manufacturers operating in the same industry as BMW and selling similar products such as passenger vehicles, SUVs, and electric cars. In selecting the peer group, we considered factors such as geographical focus in Europe, comparable market presence, product overlap, and common industry challenges including supply chain pressure and the transition to electric mobility. They form a highly relevant benchmark group for comparison

Methods and Ratio Used

Activity Ratios: To evaluate operational efficiency, we calculated inventory turnover, total asset turnover, and fixed asset turnover. These ratios were chosen to assess how effectively the companies manage inventory and utilize both total and fixed assets to generate sales. This is particularly important in the automotive industry, which is highly capital-intensive and relies heavily on efficient asset utilization.

Profitability Ratios: Profitability was assessed using gross profit margin, operating profit margin, return on assets (ROA), and return on equity (ROE). These ratios were selected to measure how well the companies convert revenue into profit, how efficiently assets are used to generate earnings, and the level of returns delivered to shareholders.

Liquidity Ratios: Liquidity was analysed using the current ratio, quick ratio, and cash ratio. These ratios were chosen to examine the companies' ability to meet short-term obligations under different liquidity conditions, ranging from overall current assets to the most conservative measure based solely on cash and cash equivalents.

Solvency Ratios: To assess long-term financial stability, we calculated the debt-to-equity ratio, debt-to-asset ratio, equity ratio, and interest coverage ratio. These ratios were selected to evaluate capital structure, reliance on debt financing, and the ability of the companies to service interest obligations from operating earnings.

Challenge: The main challenges were structuring the data effectively for clear visualizations, particularly deciding which variables should be placed in rows or columns. Choosing appropriate colours was also difficult, as colour selection strongly influences both visual appeal and interpretability. Additionally, balancing the presentation of key insights while avoiding visual clutter, especially when working with multiple ratios and categories, to ensure the dashboards remained clear and easy to understand.

Interpretation of Results

Activity Ratio: The activity ratios position BMW very strongly relative to its peer group, indicating high operational efficiency. BMW records consistently **higher inventory turnover**, rising from about **5.43 in 2018** to a peak of **6.20 in 2022**, before normalising to around **5.41 in 2023**, showing fast inventory movement and strong demand. In terms of asset utilisation, **BMW's total asset turnover** improves from approximately **0.48 in 2018** to about **0.62 in 2023**, reflecting more efficient use of its asset base to generate sales. **Fixed asset turnover** also strengthens over time, increasing from roughly **1.30 in 2018** to around **1.50 in 2023**, indicating effective use of production facilities and equipment. Compared to peers, BMW's higher and more stable activity ratios highlight a clear operational advantage, which can be explained by efficient production planning, disciplined asset management, and a strong premium-oriented business model.

Profitability Ratio: The profitability ratios position BMW very strongly relative to its peer group, indicating superior profit generation and returns. BMW records consistently higher **gross profit margins**, ranging roughly between **24%** and **26%** across most years, indicating strong pricing power and cost control. Its **operating profit margin** also outperforms peers, improving from about **9.3% in 2018** to nearly **11.9% in 2023**, showing efficient management of operating costs. BMW also recorded the highest **net margins**, rising from **7% in 2018** to **13% in 2022**, reflecting strong cost control and premium pricing. In terms of asset profitability, BMW's **return on assets (ROA)** rises from approximately **3.5% in 2018** to a peak of about **7.8% in 2022** but decrease to **4.9% in 2023**, before moderating to around **4.9% in 2023**, indicating effective use of assets to generate profits. Similarly, **return on equity**

(ROE) remains strong, increasing from roughly **12.7% in 2018** to over **22% in 2022** but decreased to **13.21% in 2023**, highlighting attractive returns to shareholders. Compared to peers, BMW's higher profitability reflects its premium positioning, strong brand demand, and disciplined cost and capital management, while weaker peer performance can be explained by thinner margins and lower returns in more volume-focused business models.

Liquidity Ratio: The liquidity ratios show that BMW maintains a balanced and sufficient short-term liquidity position relative to its peer group. BMW's **current ratio** remains stable around **1.18** and **1.09** over the period, indicating that current assets are adequate to cover short-term liabilities without holding excess idle resources. Its **quick ratio** stays close to **0.98** and **0.81**, which is solid for the automotive industry and suggests that BMW can meet short-term obligations without relying heavily on inventory sales. The **cash ratio** ranges approximately between **0.15** and **0.20**, reflecting a disciplined cash management approach rather than excessive cash holdings. Compared to peers, BMW does not have the highest liquidity buffers, but its ratios indicate efficient working-capital management and controlled liquidity risk, which is appropriate for a capital-intensive manufacturer with stable cash flows.

Solvency Ratio: The solvency ratios position BMW very strongly relative to its peer group, indicating low financial risk and a solid long-term financial structure. BMW's **debt-to-equity ratio** declines steadily from around **1.76 in 2018** to approximately **0.98 in 2023**, showing reduced reliance on debt financing over time. Similarly, the **debt-to-asset ratio** falls from about **0.49** to roughly **0.36**, while the **equity ratio** increases from around **0.27 in 2018** to about **0.37 in 2023**, reflecting a strengthening equity base. In addition, BMW's **interest coverage ratio** improves significantly from **18.8 in 2020**, reaching very high levels by **97.7 in 2023**, which indicates that operating earnings comfortably cover interest expenses. Compared to peers, these results highlight BMW's strong balance sheet, conservative leverage, and high capacity to service debt, which can be explained by disciplined financial management and stable operating cash flows.

Investment Recommendation: Overall, BMW appears financially stronger than its competitors, showing high operational efficiency, strong profitability, balanced liquidity, and low financial risk. Despite not having the highest liquidity buffers, its liquidity is well managed and appropriate for a capital-intensive industry. Based on this strong financial position, BMW is well placed to handle industry challenges, and investing in the company is recommended